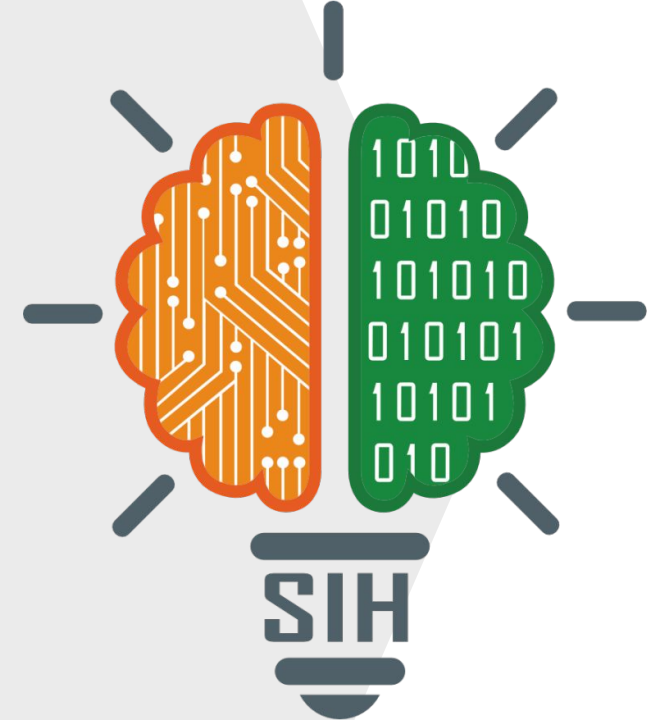


SMART INDIA HACKATHON 2025



TITLE PAGE

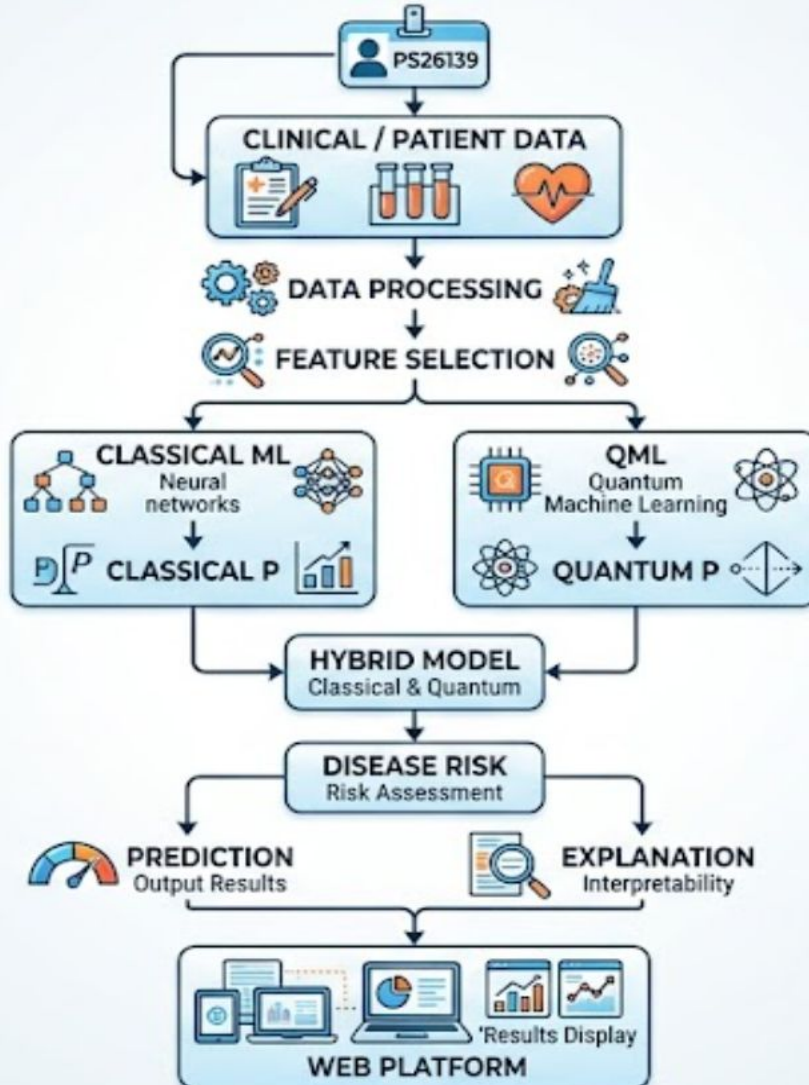
- **Problem Statement ID – 26139**
- **Problem Statement Title-Hybrid**
Quantum Machine Learning for Early
Disease Detection
- **Theme-** Quantum and AI in Healthcare
- **PS Category-** Software
- **Team ID-** SIH25XXXX



Proposed Solution

- **Unified platform** combining classical ML with Quantum Machine Learning (QML)
- Ingests **high-dimensional biomedical data** such as:
 - Genomics
 - Medical imaging
 - Electronic Health Records
- Performs **data cleaning, normalization and feature selection**
- Compresses relevant features into a **quantum-compatible representation**
- Trains hybrid models such as:
 - Quantum Support Vector Machine
 - Variational Quantum Classifier
 - Quantum Neural Network
- Generates **disease-risk predictions with confidence scores**
- Provides **explainable predictions** to assist medical interpretation

DATA-DRIVEN DISEASE RISK PREDICTION



10. TECHNOLOGY STACK

LAYER	TECHNOLOGY
Frontend	React + Tailwind
Backend	Python + FastAPI
Classical ML	scikit-learn + XGBoost
Quantum ML	Qiskit
Quantum simulator	Qiskit Aer
Explainability	SHAP + perturbation analysis
Database	PostgreSQL
API	REST
Containerization	Docker
Visualization	Recharts / Plotly
Deployment	Cloud VM/container

Why Feasible?

- **Simulator-first:** Works without requiring quantum hardware
- **NISQ-ready:** Compatible with near-term quantum devices
- **Hybrid architecture:** Classical computing handles heavy preprocessing
- **Modular:** Supports multiple QML algorithms and diseases

Challenges → Solutions

High-dimensional data → Feature selection + dimensionality reduction

Limited qubits → Encode only important features

Quantum noise → Noise-aware simulation & testing

Hardware limitations → Simulator → Hardware progression

Roadmap

Prototype → Simulator → Benchmark → Quantum Hardware

- **Earlier Disease Detection**

Identify disease patterns from complex biomedical data and support earlier clinical intervention.

- **High-Dimensional Data Analysis**

Better utilization of complex genomic, imaging and patient-record features through intelligent feature selection and QML.

- **Improved Diagnostic Support**

Optimize for clinically important metrics such as:

- **Sensitivity + Specificity + ROC-AUC**

rather than relying only on overall accuracy.

- **Explainable AI**

Provide interpretable feature contributions and prediction confidence to improve transparency.

- **Quantum-Ready Healthcare AI**

Creates an extensible framework that can evolve alongside near-term quantum computing hardware.

Quantum Machine Learning

- Schuld & Petruccione — *Supervised Learning with Quantum Computers*
- Cerezo et al. — *Variational Quantum Algorithms*
- Havlíček et al. — *Quantum-Enhanced Feature Spaces*

Frameworks

- Qiskit
- PennyLane
- Scikit-learn
- PyTorch

Research Areas

- Quantum Support Vector Machines
- Variational Quantum Classifiers
- Quantum Neural Networks
- Explainable AI for Healthcare

Benchmarking

Classical ML: XGBoost
vs
Hybrid QML: VQC